

Task: HD Task 4.0 – Proposal for a New Android App Using Llama

**App Name: Privion Rehab –
Adaptive Recovery Intelligence**

submitted to

Deakin University - School of Information Technology
for SIT305 – Mobile Application Development

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1. Introduction

Privion Rehab is an adaptive recovery support app that helps patients track subtle body signals during rehabilitation and translate them into actionable insights. Instead of relying on generic recovery plans, it uses on-device Llama AI to detect patterns in pain, movement difficulty, and fatigue over time. Its unique value lies in transforming subjective patient input into structured, real-time insights that support safer, more personalised rehabilitation.

2. Objective

The goal of Privion Rehab is to help rehabilitation patients, physiotherapy clients, and detail-focused gym users track and understand subtle body signals often ignored during recovery or training. It addresses inconsistent self-reporting by using AI-driven pattern detection to support safer, more informed decisions for both individuals and clinicians.

3. Proposed solution

3.1. Core features and user Interaction

Feature 1: Signal logging – users record pain, fatigue, instability, or movement issues through quick taps or short text/voice input during rehab or training sessions.

Feature 2: Pattern detection and feedback – the system analyses trends across sessions and provides real-time insights such as recurring pain triggers, uneven performance, or abnormal fatigue patterns.

Feature 3: Clinician summary mode – compresses user logs into concise trend reports for physiotherapists, highlighting key recovery issues and progress patterns.

User flow: User logs session → inputs body signals → AI processes patterns → receives instant feedback → optional summary generated for clinician review.

3.2. LLM integration (Llama)

What AI does: Classifies inputs, detects patterns over time, and generates concise summaries and recommendations.

How it's used in app: After each session, Llama processes user-reported signals and historical data to identify trends and provide immediate feedback or weekly summaries.

Why on-device matters: Sensitive health data remains private, responses are instant during sessions, and the app functions without internet access.

3.3. Benefits

Personalisation: Adapts to individual recovery patterns rather than relying on generic rehab plans.

Automation: Automatically detects trends and generates summaries without manual analysis.

Accessibility: Simple input methods reduce effort for users with limited mobility or technical knowledge.

Decision support: Provides clear insights that help users and clinicians make safer, data-informed adjustments.

3.4. Safety and quality controls

Prompt rules: Restrict outputs to supportive insights, avoiding medical diagnosis or treatment claims.

Refusal cases: The system declines to answer requests requiring clinical judgement or emergency advice.

Moderation: Filters ambiguous or extreme inputs to prevent misleading interpretations.

Privacy handling: All data is processed on-device with no external storage, ensuring user confidentiality.

4. Technology stack and architecture

4.1. Tech stack

Frontend: Android Studio (Kotlin, Jetpack Compose)

AI Model: Llama 3.2 (on-device)

SDK: Google AI Edge SDK (or similar on-device inference framework)

Other tools:

- Room database (local data storage)
- ViewModel (state management)
- Speech-to-text API (optional for voice input)

4.2. Architecture diagram

UI collects user inputs (pain, fatigue, notes)

ViewModel manages state and passes structured data

AI engine processes inputs and detects patterns

Output returns insights, feedback, or summaries to UI

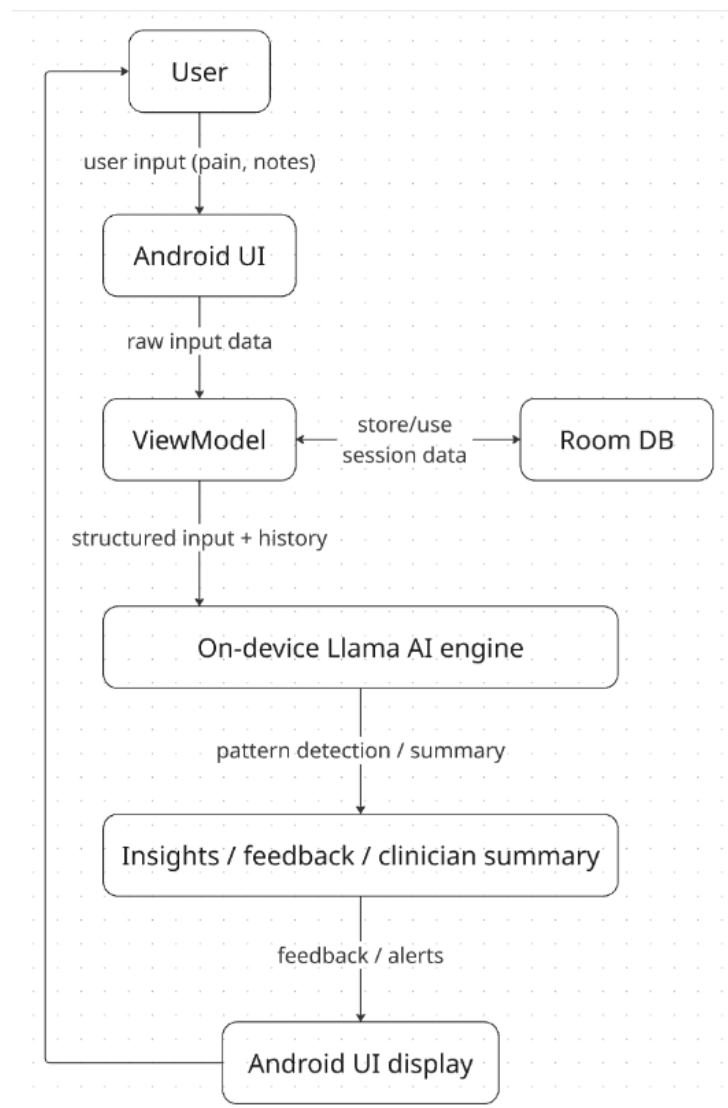


Figure 1. System architecture of Privion Rehab: on-device AI pipeline for rehabilitation signal analysis (created using Miro, 2026).

The figure shows the flow from user input through the Android UI and ViewModel to local storage and the on-device Llama engine, which processes signals and history to generate real-time feedback for adaptive recovery.

4.3. Deployment mode

On-device deployment

The application runs the Llama model locally on the user's device to ensure sensitive health and body data remains private. On-device processing also enables real-time feedback during workouts or rehabilitation sessions and allows the app to function without internet connectivity.

4.4. Feasibility

Assumption 1: Target devices support on-device AI inference with sufficient memory and processing capability.

Assumption 2: User inputs (text or voice) are frequent and consistent enough to allow meaningful pattern detection over time.

Constraint: Limited computational resources on mobile devices may restrict model size and complexity, requiring optimisation or smaller model variants.

5. Development methodology

Agile (Scrum) is adopted to support iterative development of AI-driven features in Privion Rehab. The model requires continuous testing and refinement due to variability in user input and behaviour patterns. Short development cycles enable rapid feedback, allowing adjustments to prompts, outputs, and system logic, ensuring the AI remains accurate, reliable, and aligned with real user needs.

6. Conclusion

Privion Rehab improves rehabilitation and training by turning subtle body signals into actionable insights, reducing injury risk and improving recovery awareness. Its on-device AI enables private, real-time feedback. With future expansion into predictive monitoring and clinician integration, it has strong potential to advance personalised, data-driven recovery systems.

7. LLM declaration statement

AI used: Yes

Parts assisted: Idea refinement, wording of proposal sections (introduction, objective, proposed solution, methodology, conclusion), and minor structuring of technical descriptions.

Purpose: AI was used to improve clarity, conciseness, and technical expression of the proposal. All core concepts, system design decisions, and final content were reviewed, edited, and validated to reflect my own understanding.

8. GitHub repository

Repo link: <https://github.com/triek/SIT305-Privion>

My other Web application project using the same concepts:

<https://github.com/triek/privion>

9. References

Meta AI, 2024, *Introducing Meta Llama 3: The most capable openly available LLM to date*, viewed 26 March 2026,

<https://ai.meta.com/blog/meta-llama-3/>

Edge Impulse, 2025, *What is edge machine learning (edge ML)?*, viewed 26 March 2026,

<https://docs.edgeimpulse.com/knowledge/concepts/what-is-edge-machine-learning>

Android Developers, 2025, *Guide to app architecture*, viewed 26 March 2026,

<https://developer.android.com/topic/architecture>

Android Developers, 2025, *Save data in a local database using Room*, viewed 26 March 2026,

<https://developer.android.com/training/data-storage/room>

World Health Organization, 2023, *Rehabilitation*, viewed 26 March 2026,

<https://www.who.int/news-room/fact-sheets/detail/rehabilitation>